CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

Disk

Lecture No.- 2



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Recap of Previous Lecture



Topic Magnetic Disk

Topic Disk Capacity

Topic Disk Access Time

Topics to be Covered



- Topic: Multiple Sector Access
- Topic: Disk Cylinder
- Topic: Disk Addressing

[MCQ] GATE-2011

1 rotation time = 10 ms

#Q. An application loads 100 libraries at startup. Loading each library requires exactly one disk access. The seek time of the disk to a random location is given as 10ms. Rotational speed of disk is 6000rpm. If all 100 libraries are loaded from random locations on the disk, how long does it take to load all libraries? (The time to transfer data from the disk block once the head has been positioned at the start of the block may be neglected)

(A) 0.5 s
(B) 1.25 s
(C) 1.5 s
(D) 1.0 s

$= 100 * \left(10 + \frac{10ms}{2} + 0\right)$

$= 1500$ msec

$= 1.5$ sec

Ques) Disk transfer rate = 400 MB/sec

1 sector transfer time = ——— ?

sector size = 2KB

Soln

for 400MB, time = 1 sec

for 2KB, time = $\frac{1 \text{ sec}}{400 \text{ MB}} * 2 \text{ KB}$

$= \frac{1}{200}$ msec

$= 0.005$ ms

[MCQ] GATE-2015

1 rotation time = 4 ms

#Q. Consider a typical disk that rotates at 15000 rotations per minute (RPM) and has a transfer rate of $50*10^6$ bytes/sec. If the average seek time of the disk is twice the average rotational delay and the controller's transfer time is 10 times the disk transfer time, the average time (in milliseconds) to read or write a 512-byte sector of the disk is 6.1 ?

for $50*10^6$ B, time = 1 sec

for 512 B, —11— = $\frac{1 \text{ sec}}{50*10^6} * 512$

$= \frac{512}{50}$ μsec = 10.24 μsec

= 0.01024 ms

$$= 2 * \left(\frac{4}{2}\right) + \frac{4}{2} + 0.01024 + 10 * 0.01024$$

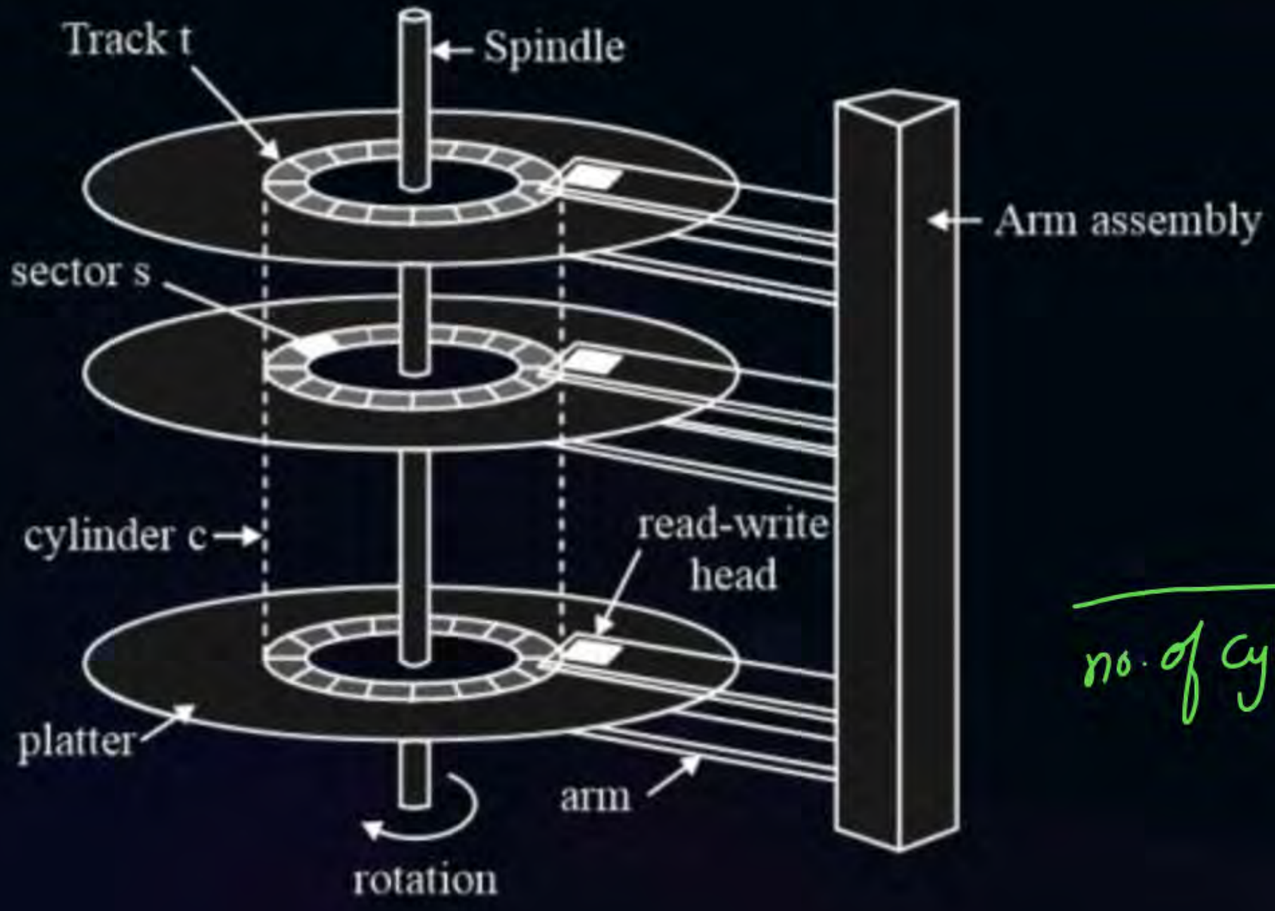
$$= 4 + 2 + 0.01024 + 0.1024$$

$$= 6.11264 \text{ ms}$$



Topic : Cylinder

To save seek time data is stored in disk cylinder wise.



cylinder :-

collection of tracks of same radius from all surface, forms a cylinder

no. of cylinders in disk = no. of tracks per surface

for one cylinder ⇒ 1 seek time

for one track ⇒ 1 rotational latency

for each sector ⇒ Transfer time needed

Ques) seek time = 10ms

1 rotation time = 20ms

No of sectors per track = 500

A file stored on 2 complete tracks

file is stored on single surface :-

$$= (2 * 10) + 2 * \left(\frac{20}{2}\right) + 1000 * \frac{20}{500}$$

$$= 80 \text{ ms}$$

file is stored cylinder wise :-

$$\text{file access time} = 10 + 2 * \left(\frac{20}{2}\right) + 1000 * \frac{20 \text{ ms}}{500}$$

$$= 10 + 20 + 40$$

$$= 70 \text{ ms}$$

no. of sectors per cylinder = no. of surfaces in disk * no. of sectors per track

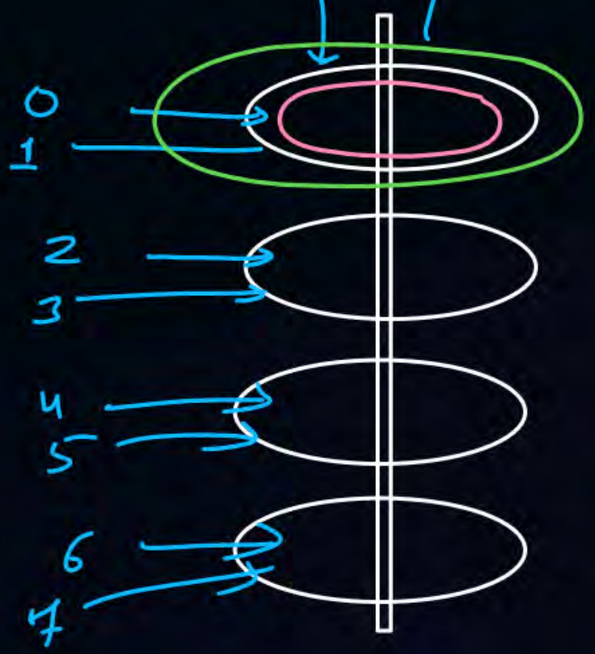
Topic : Disk Addressing



4 platters ⇒ 8 surfaces

1 track ⇒ 10 sectors (0 to 9)

no. of sectors per cylinder = 8 * 10 = 80



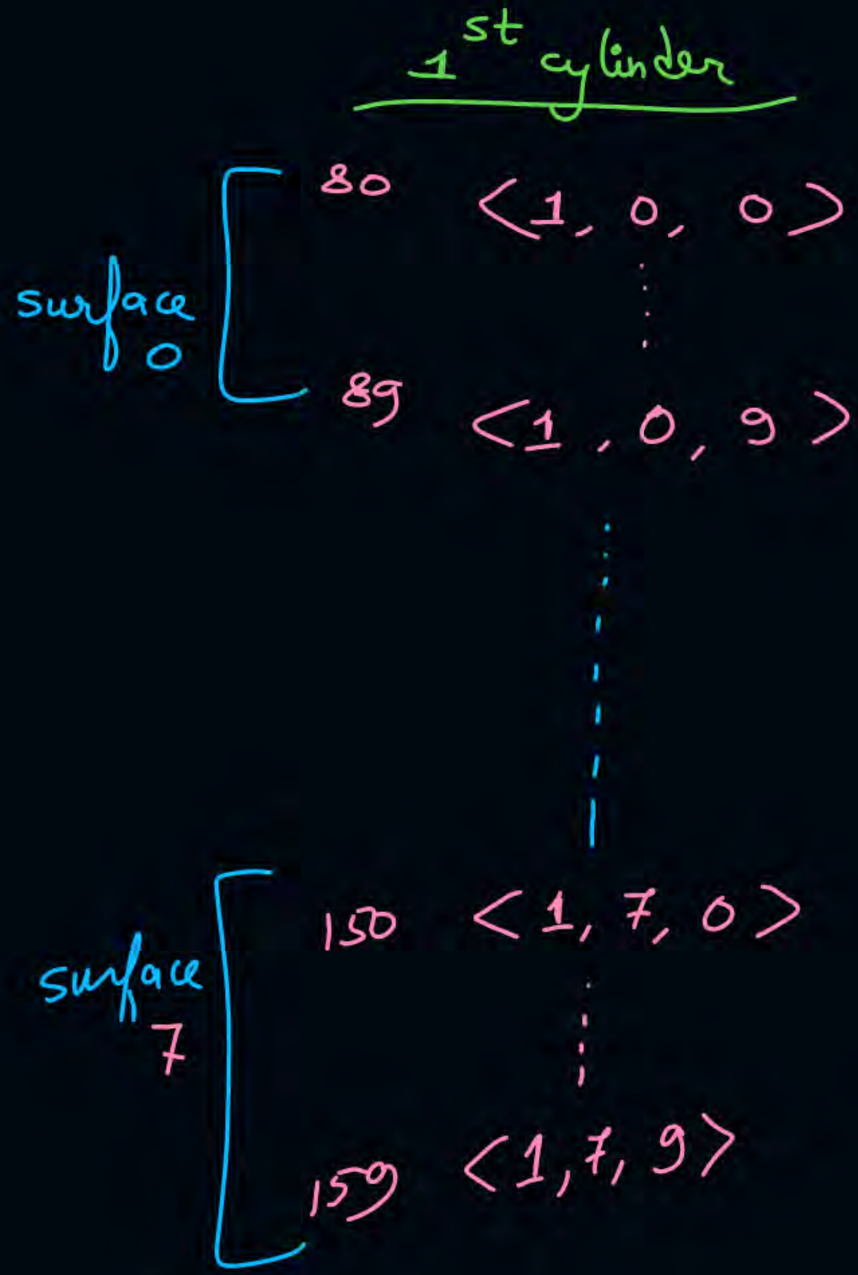
<cylinder no, surface no, sector no>

0th cylinder

sector no.	add.
surface 0: 0	<0, 0, 0>
1	<0, 0, 1>
⋮	⋮
9	<0, 0, 9>
surface 1: 10	<0, 1, 0>
11	<0, 1, 1>
⋮	⋮
19	<0, 1, 9>

surface 7

70	<0, 7, 0>
⋮	⋮
79	<0, 7, 9>



Add :- $\langle 3, 5, 8 \rangle$

$= (3 * 80) + (5 * 10) + 8$

$= 240 + 50 + 8$

$= 298$

$<c, h, s>$

$$\text{Sector no.} = (c * n_c) + (h * n_t) + s$$

n_c = no. of sectors per cylinder

n_t = no. of sectors per track

$$c = \lfloor \text{sector no.} / n_c \rfloor$$

$$h = \lfloor (\text{sector no.} \,\%\, n_c) / n_t \rfloor$$

$$s = (\text{sector no.} \,\%\, n_c) \,\%\, n_t$$

[NAT]

#Q. A hard disk has 16 sectors per track, 4 platters each with 2 recording surfaces and 32 cylinders. The address of a sector is given as a triple ⟨c,h,s⟩, where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0th sector is addressed as ⟨0,0,0⟩, the 1st sector as ⟨0,0,1⟩, and so on.

The address ⟨12, 6 ,12⟩ corresponds to sector number?

$n_t = 16$

$n_c = 8 * 16 = 128$

$= (12 * 128) + (6 * 16) + 12$

$= \underline{1644}$ Ans.

[NAT]

Ans $= \langle 6, 6, 3 \rangle$

#Q. A hard disk has 16 sectors per track, 4 platters each with 2 recording surfaces and 32 cylinders. The address of a sector is given as a triple ⟨c,h,s⟩, where c is the cylinder number, h is the surface number and s is the sector number. Thus, the 0th sector is addressed as ⟨0,0,0⟩, the 1st sector as ⟨0,0,1⟩, and so on.

The address of 867th sector?

$n_t = 16$

$n_c = 128$

$c = \lfloor 867 / 128 \rfloor = 6$

$h = \lfloor (867 \% 128) / 16 \rfloor = 6$

$s = (867 \% 128) \% 16 = 3$

[NAT]

#Q. Consider a hard disk with 36 recording surfaces (0-35) having 10000 cylinders (0-9999) and each track contains 64 sectors (0-63). Data in disk are organized cylinder-wise and the addressing format is <cylinder no., surface no., sector no.>. A file in the disk is stored starting from address <1660, 28, 38>. What is the sector number of the first sector of the file in the disk?

$n_t = 64$

$n_c = 64 * 36$

$= 2304$

$= (1660 * 2304) + (28 * 64) + 38$

$= 3826470$

[NAT]

#Q. In above questions if a file is stored on 55788 sectors in contiguous manner then what is the sector number of the last sector of the file?

$$= 3826470 + 55788 - 1$$

$$= 3882257$$

[NAT]

#Q. Calculate the address in format <c, h, s> for the last sector of the file?

$c = \lfloor 3882257 / 2304 \rfloor = 1685$

$h = \lfloor (3882257 \% 2304) / 64 \rfloor = 0$

$s = (3882257 \% 2304) \% 64 = 17$

<1685, 0, 17>

[NAT]

H.W.

#Q. Consider a hard disk with 16 recording surfaces (0-15) having 16384 cylinders (0-16383) and each track contains 64 sectors (0-63). Data storage capacity of in each sector is 512 Bytes. Data are organized cylinder-wise and addressing format is <cylinder no., surface no., sector no.>. A file of size 42797 KB is stored in the disk and the starting disk location of the file is <1200, 9, 40>. What is the cylinder number of the last sector of the file, if it is stored in a contiguous manner?



2 mins Summary

Topic Multiple Sector Access

Topic Disk Cylinder

Topic Disk Addressing



Happy Learning

THANK - YOU